

SAMAYUKTAM

Cross-Machine Determinism Verification Report

Generated: 2026-06-09 10:30 UTC

1. Executive Summary

Machines tested	4
Total sort runs	20
Successful	20
Failed	0
Files compared	5
Hash-identical files	5
Divergent files	0
Verdict	DETERMINISTIC

This report was produced by running **samayuktam_cpu** (Sort / PCMP mode) on identical .bin input files across heterogeneous hardware. Each produced .sorted.pcmp artifact is SHA-256 hashed and compared. Bit-identical hashes confirm full cross-platform determinism as guaranteed by the PCMP v1 / SPCMP v1 specifications.

2. Hash Comparison Matrix

File	19e2290be983 (cpu)	6ee456cbd193 (cpu)	Prime.localdomain (cpu)	Prime.localdomain (gpu)
test_all_zeros.bin	4a15ebcc7eae...	4a15ebcc7eae...	4a15ebcc7eae...	4a15ebcc7eae...
test_edge_nan.bin	8a4e6194326f...	8a4e6194326f...	8a4e6194326f...	8a4e6194326f...
test_large.bin	d330b42bd899...	d330b42bd899...	d330b42bd899...	d330b42bd899...
test_normal.bin	4696ee340d80...	4696ee340d80...	4696ee340d80...	4696ee340d80...
test_sorted_desc.bin	334cf11352cd...	334cf11352cd...	334cf11352cd...	334cf11352cd...

3. Per-Machine Details

Machine: 19e2290be983

Collected at: 2026-06-09T10:29:59Z Mode: CPU

Field	Value
Hostname	19e2290be983
OS	Ubuntu 22.04.5 LTS
Kernel	6.6.122+
Architecture	x86_64
CPU Model	Intel(R) Xeon(R) CPU @ 2.20GHz
Cores / Threads	2 / 2
Memory	12.67 GB
Idd Version	Idd (Ubuntu GLIBC 2.35-0ubuntu3.13) 2.35 unavailable
gcc Version	gcc (Ubuntu 11.4.0-1ubuntu1~22.04.3) 11.4.0
GPU	none
Environment	colab-cpu

Sort Runs

Input File	Status	Elapsed (ms)	Output Bytes	PCMP SHA-256
test_all_zeros.bin	success	13	149	4a15ebcc7eae413e...
test_edge_nan.bin	success	10	17882	8a4e6194326fb92c...
test_large.bin	success	89	455425	d330b42bd89915df...
test_normal.bin	success	10	19124	4696ee340d809674...
test_sorted_desc.bin	success	10	22121	334cf11352cd8936...

Machine: 6ee456cbd193

Collected at: 2026-06-09T10:25:50Z Mode: CPU

Field	Value
Hostname	6ee456cbd193
OS	Ubuntu 22.04.5 LTS
Kernel	6.6.122+
Architecture	x86_64
CPU Model	AMD EPYC 7B13
Cores / Threads	24 / 24
Memory	47.05 GB
Idd Version	Idd (Ubuntu GLIBC 2.35-0ubuntu3.13) 2.35 unavailable
gcc Version	gcc (Ubuntu 11.4.0-1ubuntu1~22.04.3) 11.4.0
GPU	none
Environment	colab-cpu

Sort Runs

Input File	Status	Elapsed (ms)	Output Bytes	PCMP SHA-256
test_all_zeros.bin	success	14	149	4a15ebcc7eae413e...
test_edge_nan.bin	success	730	17882	8a4e6194326fb92c...
test_large.bin	success	34	455425	d330b42bd89915df...
test_normal.bin	success	7	19124	4696ee340d809674...
test_sorted_desc.bin	success	6	22121	334cf11352cd8936...

Machine: Prime.localdomain

Collected at: 2026-06-09T10:25:14Z Mode: CPU

Field	Value
Hostname	Prime.localdomain
OS	Ubuntu 24.04.3 LTS
Kernel	6.6.87.2-microsoft-standard-WSL2
Architecture	x86_64
CPU Model	12th Gen Intel(R) Core(TM) i5-12450HX
Cores / Threads	12 / 12
Memory	13.65 GB
Idd Version	ldd (Ubuntu GLIBC 2.39-0ubuntu8.7) 2.39
gcc Version	gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
GPU	NVIDIA GeForce RTX 3050 6GB Laptop GPU, 595.97, 6144 MiB;
Environment	local-gpu

Sort Runs

Input File	Status	Elapsed (ms)	Output Bytes	PCMP SHA-256
test_all_zeros.bin	success	55	149	4a15ebcc7eae413e...
test_edge_nan.bin	success	73	17882	8a4e6194326fb92c...
test_large.bin	success	222	455425	d330b42bd89915df...
test_normal.bin	success	56	19124	4696ee340d809674...
test_sorted_desc.bin	success	57	22121	334cf11352cd8936...

Machine: Prime.localdomain

Collected at: 2026-06-09T10:24:46Z Mode: GPU | Booster: libsamayuktam_cuda.so

Field	Value
Hostname	Prime.localdomain
OS	Ubuntu 24.04.3 LTS
Kernel	6.6.87.2-microsoft-standard-WSL2
Architecture	x86_64
CPU Model	12th Gen Intel(R) Core(TM) i5-12450HX
Cores / Threads	12 / 12
Memory	13.65 GB
Idd Version	Idd (Ubuntu GLIBC 2.39-0ubuntu8.7) 2.39
gcc Version	gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
GPU	NVIDIA GeForce RTX 3050 6GB Laptop GPU, 595.97, 6144 MiB;
Environment	local-gpu

Sort Runs

Input File	Status	Elapsed (ms)	Output Bytes	PCMP SHA-256
test_all_zeros.bin	success	69	149	4a15ebcc7eae413e...
test_edge_nan.bin	success	467	17882	8a4e6194326fb92c...
test_large.bin	success	286	455425	d330b42bd89915df...
test_normal.bin	success	267	19124	4696ee340d809674...
test_sorted_desc.bin	success	63	22121	334cf11352cd8936...

4. Appendix — Full PCMP SHA-256 Hashes

19e2290be983

test_all_zeros.bin	4a15ebcc7eae413eeb212b2111ac7d1b005cd0e633bd96dc6d18b4100383e929
test_edge_nan.bin	8a4e6194326fb92c23f09d4f288f603381024b779b83068298382a71c0ecb07c
test_large.bin	d330b42bd89915dfef60cc1217c878974ddd79d761c57e2c774381404f9da93c
test_normal.bin	4696ee340d80967420a0fdd648e9f47870c3688a4cb7663be6a8251b3997fbb2
test_sorted_desc.bin	334cf11352cd89362f9f002054f9b0e97865d984546c56ff325904f75db7ccf0

6ee456cbd193

test_all_zeros.bin	4a15ebcc7eae413eeb212b2111ac7d1b005cd0e633bd96dc6d18b4100383e929
test_edge_nan.bin	8a4e6194326fb92c23f09d4f288f603381024b779b83068298382a71c0ecb07c
test_large.bin	d330b42bd89915dfef60cc1217c878974ddd79d761c57e2c774381404f9da93c
test_normal.bin	4696ee340d80967420a0fdd648e9f47870c3688a4cb7663be6a8251b3997fbb2

test_sorted_desc.bin

334cf11352cd89362f9f002054f9b0e97865d984546c56ff325904f75db7ccf0

Prime.localdomain

test_all_zeros.bin

4a15ebcc7eae413eeb212b2111ac7d1b005cd0e633bd96dc6d18b4100383e929

test_edge_nan.bin

8a4e6194326fb92c23f09d4f288f603381024b779b83068298382a71c0ecb07c

test_large.bin

d330b42bd89915dfef60cc1217c878974ddd79d761c57e2c774381404f9da93c

test_normal.bin

4696ee340d80967420a0fdd648e9f47870c3688a4cb7663be6a8251b3997fbb2

test_sorted_desc.bin

334cf11352cd89362f9f002054f9b0e97865d984546c56ff325904f75db7ccf0

Prime.localdomain

test_all_zeros.bin

4a15ebcc7eae413eeb212b2111ac7d1b005cd0e633bd96dc6d18b4100383e929

test_edge_nan.bin

8a4e6194326fb92c23f09d4f288f603381024b779b83068298382a71c0ecb07c

test_large.bin

d330b42bd89915dfef60cc1217c878974ddd79d761c57e2c774381404f9da93c

test_normal.bin

4696ee340d80967420a0fdd648e9f47870c3688a4cb7663be6a8251b3997fbb2

test_sorted_desc.bin

334cf11352cd89362f9f002054f9b0e97865d984546c56ff325904f75db7ccf0